

GALVESTON BUOY 42035, TX, USA

WMO: 996970

Lat: 29.232N

Lon: 94.413W

Elev: 0

StdP: 14.70

Time Zone: -6.00 (W06)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.7	41.4	24.8	18.9	43.7	28.8	22.8	46.5	32.8	46.6	31.3	47.1	19.0	350	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	87.1	78.9	86.4	78.7	86.0	78.6	81.3	85.0	80.6	84.6	80.2	84.5	8.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	157.0	84.1	79.4	152.8	83.8	78.9	149.8	83.7	44.8	85.0	44.2	84.8	43.7	84.6	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.4	25.7	23.3	DB	34.7	88.6	3.9	1.4	31.9	89.6	29.6	90.4	27.4	91.1	24.6	92.1	
			WB	31.0	81.9	3.8	2.0	28.3	83.3	26.0	84.5	23.9	85.6	21.2	87.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.7	54.7	56.2	62.4	69.0	75.8	81.7	83.1	83.9	81.0	75.0	65.9	58.4
	DBStd	11.43	6.32	6.44	5.56	4.31	3.38	1.98	1.72	1.81	2.52	4.79	6.25	6.57
	HDD50	73	32	23	4	0	0	0	0	0	0	0	2	12
	HDD65	977	318	248	110	15	1	0	0	0	0	3	64	218
	CDD50	7613	179	195	388	571	800	950	1026	1051	929	774	478	272
	CDD65	3044	0	1	29	137	336	500	561	586	479	312	90	13
	CDH74	29683	0	0	2	119	1987	5685	6985	7555	5236	2049	65	0
	CDH80	8894	0	0	0	2	39	1556	2590	3148	1408	151	0	0
Wind	WSAvg	12.9	13.5	13.4	12.5	13.9	13.9	11.8	11.4	10.5	12.6	13.5	14.0	13.6
Precipitation	PrecAvg	56.6	4.8	3.4	3.8	3.9	4.9	5.5	5.4	4.9	5.9	5.2	4.1	4.7
	PrecMax	78.8	10.9	7.6	8.4	9.0	9.4	16.2	14.1	10.6	14.1	15.6	10.6	10.2
	PrecMin	29.8	0.6	1.0	0.7	0.4	0.0	0.6	1.3	0.7	1.4	0.5	0.7	1.3
	PrecStd	11.9	2.6	2.1	2.0	2.6	2.8	3.4	3.1	3.1	2.8	4.5	2.4	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.1	68.2	73.2	78.4	82.7	87.3	87.1	88.8	86.4	83.8	77.1	72.2
		MCWB	65.5	67.3	70.8	74.3	74.7	79.4	78.5	79.7	77.8	77.5	73.7	70.6
	2%	DB	64.8	66.3	71.5	76.5	81.0	85.8	86.3	87.4	85.5	82.7	75.7	70.6
		MCWB	64.1	65.4	70.2	73.5	76.0	78.1	78.6	79.0	78.0	76.9	71.7	69.0
	5%	DB	63.8	65.2	70.2	75.4	80.3	84.9	85.7	86.9	85.1	81.9	74.7	69.2
		MCWB	63.0	64.3	68.8	73.1	75.8	77.7	78.6	78.9	77.8	76.4	71.0	67.5
	10%	DB	62.5	63.9	69.0	74.2	79.8	84.3	85.3	86.4	84.5	80.7	73.4	67.7
		MCWB	61.4	62.6	67.4	71.8	75.7	77.4	78.6	78.8	77.4	75.5	70.3	66.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	68.2	72.3	76.2	79.6	82.3	81.4	82.4	81.9	80.3	75.5	71.2
		MCDB	66.1	68.2	72.4	76.8	79.7	84.9	85.1	86.1	83.7	81.3	77.1	71.7
	2%	WB	64.5	66.1	70.9	75.1	78.6	80.7	80.5	81.4	80.4	79.1	73.3	69.6
		MCDB	64.6	66.2	71.2	75.7	79.3	83.6	84.7	85.7	83.8	80.4	74.4	70.3
	5%	WB	63.5	64.8	69.5	74.0	77.8	79.9	80.3	80.8	79.8	78.1	72.4	68.3
		MCDB	63.6	65.0	70.0	74.8	79.0	83.3	84.6	85.5	83.8	80.2	73.8	69.3
	10%	WB	62.1	63.3	68.3	72.9	77.0	79.1	79.9	80.3	79.0	77.0	71.3	66.4
		MCDB	62.5	63.7	68.9	73.7	78.7	83.0	84.4	85.3	83.5	79.9	73.0	67.4
Mean Daily Temperature Range	5% DB	MDBR	6.7	6.4	5.0	4.1	3.2	3.0	3.2	3.4	4.0	4.8	6.1	6.8
		MCDBR	6.4	5.6	3.9	4.3	3.1	3.2	3.0	3.9	3.9	4.4	5.7	6.0
		MCWBR	7.6	6.1	4.5	5.2	3.6	3.4	2.9	3.4	3.4	4.6	7.4	7.4
	5% WB	MCDBR	6.0	5.2	3.5	4.1	2.8	3.0	3.1	3.3	4.0	4.4	5.8	6.1
		MCWBR	7.5	6.2	4.4	5.1	3.5	3.4	3.3	3.5	4.0	6.0	7.5	7.5
Clear-Sky Solar Irradiance	taub	0.352	0.365	0.380	0.431	0.475	0.461	0.483	0.466	0.446	0.389	0.375	0.357	
	taud	2.521	2.494	2.441	2.299	2.211	2.315	2.274	2.324	2.337	2.484	2.466	2.512	
	Ebn at Noon	281	286	288	275	262	264	258	262	264	274	269	273	
	Edh at Noon	29	32	35	42	46	41	43	41	39	31	30	28	
All-Sky Solar Radiation	RadAvg	930	1063	1471	1763	2009	2066	1938	1848	1595	1423	1082	830	
	RadStd	92	171	127	135	125	142	169	142	167	167	108	93	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+214	+175

Nomenclature: See separate page